Mathematical Basics — Cheat Sheet

Probabilistic Machine Learning

WiSe 2025

This sheet summarizes some essential background material you may have seen during your undergraduate studies. We will rely on these rules and techniques throughout the course. Most sections end with a couple of exercises for you to test your understanding.

1. Expectation and Integral Rules

Facts

1. Expectation of a function $g(X)$: $\mathbb{E}[g(X)] = \int g(x)p(x)\,dx$, where $p(x)$ is the *probability density function* of X.
2. Linearity: $\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$.
3. If X and Y are independent: $\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$.
4. Variance: $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Exercises

(a) Let $X \sim \mathrm{Uniform}(0, 1)$. Compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

(b) Show that $\mathbb{E}[a + bX] = a + b\,\mathbb{E}[X]$.

2. Integration by Substitution

One-dimensional reminder

If $u = g(x)$ is differentiable and g is invertible,

$$\int f(g(x))\,g'(x)\,dx = \int f(u)\,du.$$

Multidimensional change of variables

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a C^1-diffeomorphism with inverse T^{-1}. For any integrable function f,

$$\int_{T(U)} f(\mathbf{y})\,d\mathbf{y} \;=\; \int_U f\big(T(\mathbf{x})\big)\,|\det J_T(\mathbf{x})|\,d\mathbf{x},$$

where $J_T(\mathbf{x}) = \big[\partial T_i/\partial x_j\big]_{i,j}$ is the Jacobian matrix of T at $\mathbf{x}$ and $|\det J_T|$ is the absolute value of its determinant.

Remarks

- $|\det J_T(\mathbf{x})|$ describes how volumes are scaled locally by T.
- For a linear map $T(\mathbf{x}) = A\mathbf{x}$ the Jacobian is constant: $J_T = A$ and $|\det J_T| = |\det A|$.

3. Logarithm and Exponential Rules

Facts

1. log is the inverse function of exp and vice-versa: $\log(e^x) = x$ for all $x \in \mathbb{R}$, and $e^{\log x} = x$ for all $x > 0$.
2. log is *monotonically increasing* or *order-preserving*: $\log a \geq \log b$ whenever $a \geq b$.
3. $\log(ab) = \log a + \log b, \quad \log(a/b) = \log a - \log b.$
4. $\log(a^r) = r \log a.$
5. $e^{a+b} = e^a e^b, \quad \frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} \log x = 1/x.$

Exercises

Use only the facts stated above.

(a) Simplify $\log\left(\frac{e^x}{e^y}\right)$.

(b) Show that $\log(1 + e^x)$ is always positive.

(c) Show that exp is also monotonically increasing.

4. Matrix Algebra & Calculus

Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a function mapping vectors to scalars. If it is differentiable at a point $x = (x_1, \ldots, x_n)^T$, the gradient of f at that point is defined by

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x). \end{pmatrix}$$

We use subscripts, e.g. ∇_x, to denote the gradient with respect to certain variables only.

Facts

Let A, B, C and D be matrices of compatible shape.

1. 2-norm of a vector: $|x|_2^2 = x^T x.$
2. Matrix multiplication is associative: $(AB)C = A(BC).$
3. Matrix multiplication is not commutative: $AB \neq BA$ in general.
4. Matrix multiplication is distributive: $(A + B)(C + D) = AC + AD + BC + BD.$
5. Transposition reverses order: $(AB)^T = B^T A^T$
6. Gradient of linear form: $\nabla_x(b^T x) = b.$
7. Gradient of quadratic form: $\nabla_x(x^T A x) = (A + A^T)x.$

Exercises

(a) Show the fact 6 (gradient of linear form) by hand.

(b) Also show fact 7 by hand.

(c) Compute $\nabla_x(b^T A x)$

(d) Compute $\nabla_x(x^T x)$.

(e) Let A be symmetric. Compute $\nabla_x(x^T A x)$.

(f) Show that $|x - Ay|_2^2 = |x|_2^2 - 2x^T A y + |Ay|_2^2.$

Exercises 1

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October 18, 2024

1 Probability theory

Exercise 1. Let Ω be a set. Prove that $\mathcal{A} = \{\Omega, \varnothing\}$ is a σ-algebra.

Exercise 2. Let Ω be a finite set. Prove that the set $\mathcal{A}$ containing all subsets of Ω is a σ-algebra.

Exercise 3. Let Ω be a set and $\mathcal{A}$ and $\mathcal{B}$ are two σ-algebras on Ω. Show that $\mathcal{A} \cap \mathcal{B}$ is a σ-algebra on Ω.

Exercise 4. Let E and F be two sets, with $\mathcal{E}$ a σ-algebra on E. Let $X : E \to F$ an application. Show that:

$$\mathcal{F} := \{B \subseteq F,\ X^{-1}(B) \in \mathcal{E}\}$$

is a σ-algebra on F.

Exercise 5. Let $\mathbb{P}$ be a probability on the space $(\Omega, \mathcal{A})$ (as defined in the lecture). Prove the following properties:

- $\mathbb{P}(\varnothing) = 0$
- $\mathbb{P}(A) + \mathbb{P}(A^c) = 1$
- $\mathbb{P}(A \cup B) + \mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B)$
- $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$

Exercise 6. Let Ω be a finite set: $\Omega = \{w_1, \dots, w_n\}$. Let $\mathcal{A}$ be the σ-algebra defined in Exercise 2. Let $(p_1, \dots, p_n)$ such that $\sum_{i=1}^n p_i = 1$ and $p_i \geq 0$ for all i. Show that there exists a unique probability $\mathbb{P}$ on $(\Omega, \mathcal{A})$ such that for all i, $\mathbb{P}(\{w_i\}) = p_i$.

Exercise 7. Let $(\Omega, \mathcal{A})$ be a probability space. If X is a random variable on E and Y a random variable on F, show that (X, Y) is a random variable on $E \times F$.

2 Important distributions

We remind that a real random variable X has a probability density function f (where f is non-negative and measurable) if for all $x \in \mathbb{R}$:

$$\mathbb{P}_X((-\infty, x]) = \int_{-\infty}^{x} f(y)\, dy$$

In the following, we will use the notation $\mathbb{P}(X \leq x)$.

We define the expected value of X (denoted $\mathbb{E}(X)$) by:

$$\mathbb{E}(X) = \int_{\Omega} X(\omega)\mathbb{P}(d\omega) = \int_{x \in \mathbb{R}} x\, f(x)\, dx$$

Exercise 8. Show that for all x, $\mathbb{P}(X = x) = 0$.

Exercise 9. Show that if f is the probability density function for X, then $\int_{\mathbb{R}} f(x)dx = 1$.

Exercise 10. The uniform variable U on $[a, b]$ has density:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if} a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Compute $\mathbb{E}(U)$

Exercise 11. Compute the expected value for random variables following Poisson, normal and Gamma distributions.

Exercises 2

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October 24, 2024

1 Maximum Likelihood Estimation

Exercise 1. Given observations $\{x_1, \dots, x_N\}$ and a fixed value of μ, compute the maximum likelihood estimator of σ^2 for a normal distribution.

Exercise 2. Given observations $\{x_1, \dots, x_N\}$ and a fixed value of α, compute the maximum likelihood estimator of α for a Gamma distribution.

Remark: Computing the MLE estimator for α is more complicated, since it involves the derivative of the logarithm of the gamma function, called *digamma function*.

Exercise 3. Given observations $\{x_1, \dots, x_N\}$, compute the maximum likelihood estimator of λ for an exponential distribution.

Exercise 4. Given observations $\{x_1, \dots, x_N\}$, compute the maximum likelihood estimator of μ and of Σ for a multivariate normal distribution. Do not consider each component individually, but treat these quantities as vectors and use the tools of linear algebra to find the value for which the gradient vanishes.

2 Conjugate priors

Exercise 5. The geometric distribution measures the probability that the first positive value of a Bernoulli variable is observed after k draws. It writes:

$$p(k|\theta) = (1-\theta)^{k-1}\theta$$

Verify that the Beta distribution is a conjugate prior for the geometric distribution.

Exercise 6. Verify that the Gamma distribution is a conjugate prior for β in the Gamma distribution where α is known.

Exercise 7 (Important)**.** The exponential family $\mathcal{F}$ is the class of probability distributions with PDF of the form:

$$p(x|\theta) = h(x) \exp\left[\eta(\theta)^T T(x)\right]$$

where $h(.)$ is a non-negative function, $T(.)$ is called sufficient statistics, and $\eta(.)$ the natural parameter.

(a) Verify that the exponential distribution, the Poisson distribution and the normal distribution are elements of the exponential family.

(b) Write the likelihood of N i.i.d. observations $\{x_1, \ldots, x_N\}$ for a distribution of the exponential family.

(c) Show that the given definition of the exponential family is equivalent to the family of distributions of the form:

$$p(x|\theta) = f(x)g(\theta)\exp\left[\phi(\theta)^T u(x)\right]$$

(d) Using this equivalent form, show that distributions of the form

$$p(\theta) \propto g(\theta)^\eta e^{\phi(\theta)^T \nu}$$

are conjugate priors for the distributions of the exponential family.

(e) Calling $Z(\eta, \nu)$ the normalizing factor of the prior distribution (such that $p(\theta) = Z(\eta, \nu)g(\theta)^\eta e^{\phi(\theta)^T \nu}$, compute the prior predictive checking distribution associated to any distribution of the exponential family. We remind that the prior predictive distribution is given by:

$$p(x) = \int_\theta p(\theta)p(x|\theta)\, d\theta$$

Predictive checking :

* Likelihood : $p(x|\mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.

* Model distribution : $p(\mu) = \frac{1}{\sqrt{2\pi s^2}} \exp\left(-\frac{(\mu-m)^2}{2s^2}\right)$.
(prior or posterior)

Predictive checking distribution :

$$p(x) = \int_\mu p(x|\mu)\, p(\mu)\, d\mu$$

$$\propto \int_\mu \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \exp\left(-\frac{(\mu-m)^2}{2s^2}\right) d\mu$$

$$\propto \int_\mu \exp\left[-\frac{1}{2\sigma^2}\left(\mu^2 - 2x\mu + x^2\right) - \frac{1}{2s^2}\left(\mu^2 - 2m\mu + m^2\right)\right] d\mu.$$

$$\propto \int_\mu \exp\left[-\frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\mu^2 - 2\left(-\frac{x}{2\sigma^2} - \frac{m}{2s^2}\right)\mu - \frac{x^2}{2\sigma^2} - \frac{m^2}{2s^2}\right] d\mu$$

$$\propto \int_\mu \exp\left[-\frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\left(\mu^2 - 2\mu \underbrace{\frac{-\frac{x}{2\sigma^2} - \frac{m}{2s^2}}{-\frac{1}{2\sigma^2} + \frac{1}{2s^2}}}_{= \frac{s^2 x + \sigma^2 m}{\sigma^2 + s^2}}\right) - \frac{x^2}{2\sigma^2} - \frac{m^2}{2s^2}\right] d\mu$$

$$\propto \int_\mu \exp\left[-\frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\left(\mu - \frac{s^2 x + \sigma^2 m}{\sigma^2 + s^2}\right)^2 + \frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\left(\frac{s^2 x + \sigma^2 m}{\sigma^2 + s^2}\right)^2 - \frac{x^2}{2\sigma^2} - \frac{m^2}{2s^2}\right]$$

$$\propto \int_\mu \underbrace{\exp\left[-\frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\left(\mu - \frac{s^2 x + \sigma^2 m}{\sigma^2 + s^2}\right)^2\right]}_{\text{Normal PDF.}} \underbrace{\exp\left[\frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\left(\frac{s^2 x + \sigma^2 m}{\sigma^2 + s^2}\right)^2 - \frac{x^2}{2\sigma^2} - \frac{m^2}{2s^2}\right]}_{\text{Does not depend on } \mu.} d\mu$$

$$\propto \exp\left(\frac{1}{2}\left[\frac{1}{\sigma^2}+\frac{1}{s^2}\right]\left(\frac{s^2 x + \sigma^2 m}{\sigma^2 + s^2}\right)^2 - \frac{x^2}{2\sigma^2} - \frac{m^2}{2s^2}\right) \cdot \underbrace{\int_\mu \mathcal{N}(\mu; -, -)\, d\mu}_{=1}$$

Consequently:

$$p(x) \propto \exp\left[\frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\left(\frac{s^2x+\sigma^2 m}{\sigma^2+s^2}\right)^2 - \frac{x^2}{2\sigma^2} - \frac{m^2}{2s^2}\right]$$

$$\propto \exp\left[\frac{1}{2}\left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\cdot\frac{1}{(\sigma^2+s^2)^2}\left(s^4x^2+2s^2\sigma^2 m x+\sigma^4 m^2\right) - \frac{x^2}{2\sigma^2} - \frac{m^2}{2s^2}\right]$$

$$\propto \exp\left[-\frac{1}{2}\underbrace{\left(\frac{1}{\sigma^2} - \frac{\frac{1}{\sigma^2}+\frac{1}{s^2}}{(\sigma^2+s^2)^2}s^4\right)}x^2 + \left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\cdot\frac{s^2\sigma^2}{(\sigma^2+s^2)^2}\, m x\right]$$

$$= \frac{(\sigma^2+s^2)^2 - s^4 - \sigma^2 s^2}{\sigma^2(s^2+\sigma^2)^2} = \frac{\sigma^4+\sigma^2 s^2}{\sigma^2(s^2+\sigma^2)^2} = \frac{\sigma^2+s^2}{(\sigma^2+s^2)^2}$$

$$\propto \exp\left[-\frac{1}{2(s^2+\sigma^2)}x^2 + \left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\cdot\frac{s^2\sigma^2}{(\sigma^2+s^2)^2}\, m x\right]$$

$$\propto \exp\left[-\frac{1}{2(s^2+\sigma^2)}\left(x^2 - 2\left(\frac{1}{\sigma^2}+\frac{1}{s^2}\right)\cdot\frac{s^2\sigma^2}{\sigma^2+s^2}\, m x\right)\right]$$

$$\propto \exp\left[-\frac{1}{2(s^2+\sigma^2)}(x-m)^2\right]$$

$$\Rightarrow \boxed{p(x) = \mathcal{N}(x;\, m,\, \sigma^2+s^2)}.$$

Exercises 3

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1 Linear regression

Exercise 1. We consider the discriminative model for linear regression where data $\mathcal{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}$ are i.i.d. with

$$p(y_n|\mathbf{w}, x_n) = \mathcal{N}(x_n; \mathbf{w}^T x_n, \sigma^2).$$

We assume that σ^2 is fixed and known and consider a normal prior for $\mathbf{w}$: $p(\mathbf{w}) = \mathcal{N}(0, \Sigma)$. Show that the posterior $p(\mathbf{w}|\mathbf{X}, \mathbf{y})$ is a normal distribution. Give its mean μ_N and covariance matrix Σ_N.

Exercise 2. We use the same model as in the previous exercise, but we now assume that σ^2 is a variable. Show that the prior defined as $p(\mathbf{w}, \sigma^2) = p(\sigma^2)p(\mathbf{w}|\sigma^2)$ with:

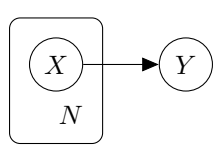
$$\begin{aligned} p(\sigma^2) &= \frac{\beta^\alpha}{\Gamma(\alpha)}(\sigma^2)^{-\alpha-1} \exp\left(-\frac{\beta}{\sigma^2}\right) \\ p(\mathbf{w}|\sigma^2) &= \mathcal{N}(\mathbf{w}; \mathbf{m}_0, \sigma^2 \mathbf{S}_0) \end{aligned}$$

Show that this prior is the conjugate for the likelihood of the Bayesian regression model.

2 Directed graphical models

Exercise 3. Prove the edge reversal property: If $G = (V, E)$ is a DAG and $(i, j) \in E$ such that $\pi_i = \varnothing$ and $\pi_j = \{i\}$, then the graph G' obtained by reversing the edge (i, j) into (j, i) is Markov equivalent to G.

Exercise 4. Consider the following graphical model:



Assuming that $p(Y|X_i) = p(Y|X_j)$ for all (i, j), show that this model corresponds to $(X_1, \dots, X_N)$ i.i.d.

Exercise 5. Consider $N+1$ random variables $\mathbf{X} = (X_0, \ldots, X_N)$ such that

$$p(\mathbf{x}) = p(x_0) \prod_{n=1}^{N} p(x_n | x_{n-1}).$$

Show that for any $i < j < k$, we have $p(x_k | x_i, x_j) = p(x_k | x_j)$.

Probabilistic Machine Learning: Exercise Sheet 02
Answers to Question 7

January 10, 2025

Question 7

(a)

- The exponential distribution has the PDF $p(x) = \lambda e^{-\lambda x}$. Setting $h(x) = 1$, $\eta(\lambda) = (\lambda, \log\lambda)^\top$ and $T(x) = (-x, 1)^\top$, we see that
$$h(x)\exp[\eta(\lambda)^\top T(x)] = 1 \cdot \exp(\lambda \cdot (-x) + \log\lambda \cdot 1) = e^{-\lambda x}\lambda.$$
- The Poisson distribution has PMF $p(x = k) = \frac{\lambda^k e^{-\lambda}}{k!}$. Setting $h(x) = \frac{1}{x!}$, $\eta(\lambda) = (\lambda, \log\lambda)^\top$ and $T(x) = (-1, x)^\top$, we see that
$$h(x)\exp[\eta(\lambda)^\top T(x)] = \frac{1}{x!}\exp[-\lambda + x\log\lambda] = \frac{\lambda^x}{x!}e^{-\lambda x}.$$
- The normal distribution has PDF $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$. By rearranging, we get
$$\begin{aligned} p(x) &= \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = \frac{1}{\sqrt{2\pi}}\exp(-\log(\sigma))\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\ &= \frac{1}{\sqrt{2\pi}}\exp\left(-x^2\frac{1}{2\sigma^2} + x\frac{\mu}{\sigma^2} - \frac{\mu}{\sigma^2} - \log\sigma\right), \end{aligned}$$
so that we have an exponential distribution with $h(x) = (2\pi)^{-1/2}$, $\eta(\mu, \sigma) = (-\frac{1}{2\sigma^2}, \frac{\mu}{\sigma^2}, -\frac{\mu^2}{\sigma^2}, -\log\sigma)^\top$ and $T(x) = (x^2, x, 1, 1)^\top$.

(b)

$$p(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^n h(x_i)\exp\left(\eta(\theta)^\top \sum_{i=1}^n T(x_i)\right)$$

(c)

The only difference between the two definitions is the θ-dependent term $g(\theta)$ in front of the exponential. We can "simulate" it in the first definition by exploiting that $g(\theta) = \exp(\log g(\theta))$, so that we can extend η and T by entries $\log g(\theta)$ and 1, respectively.

(d)

Let the prior parameter distribution be given by $p(\theta) = Cg(\theta)^\eta \exp(\phi(\theta)^\top \nu)$ for some constant C and let the likelihood be $p(x \mid \theta) = f(x)g(\theta)\exp(\phi(\theta)^\top u(x))$ Then,
$$\begin{aligned} p(x, \theta) = p(\theta)p(x \mid \theta) &= Cg(\theta)^\eta \exp(\phi(\theta)^\top \nu) f(x) g(\theta)\exp(\phi(\theta)^\top u(x)) \\ &= Cf(x)g(\theta)^{\eta+1}\exp(\phi(\theta)^\top(\nu + u(x))) \\ &\propto g(\theta)^{\eta+1}\exp(\phi(\theta)^\top(\nu + u(x))). \end{aligned}$$
In general (after n i.i.d. observations), the new parameters will be $\widetilde{\eta} = \eta + n$ and $\widetilde{\nu} = \nu + \sum_{i=1}^n u(x_i)$.

(e)

Given that $p(\theta) = Z(\eta, \nu) g(\theta)^{\eta} \exp(\phi(\theta)^{\top} \nu)$ and $p(x \mid \theta) = f(x) g(\theta) \exp(\phi(\theta)^{\top} u(x))$, we have

$$\begin{aligned} p(x) &= \int_{\theta} p(\theta) p(x \mid \theta) \mathrm{d}\theta = Z(\eta, \nu) f(x) \int_{\theta} g(\theta)^{\eta+1} \exp(\phi(\theta)^{\top} (\nu + u(x))) \mathrm{d}\theta \\ &= f(x) \frac{Z(\eta, \nu)}{Z(\eta + 1, \nu + u(x))}. \end{aligned}$$

The last equality is true, because the term under the integral is simply an unnormalized PDF, so its result must be the inverse of its normalization factor.